

BUILDING BLOCKS

Building blocks for dark software factories

From conversation to validated software

CONVERSATION → CONTRACTS → EVIDENCE → RELEASE

Andreas Petersson

Autonomy requires control

UNATTENDED

≠ OPAQUE

Work may continue without a person at every step.

State, evidence and intervention points must remain visible.

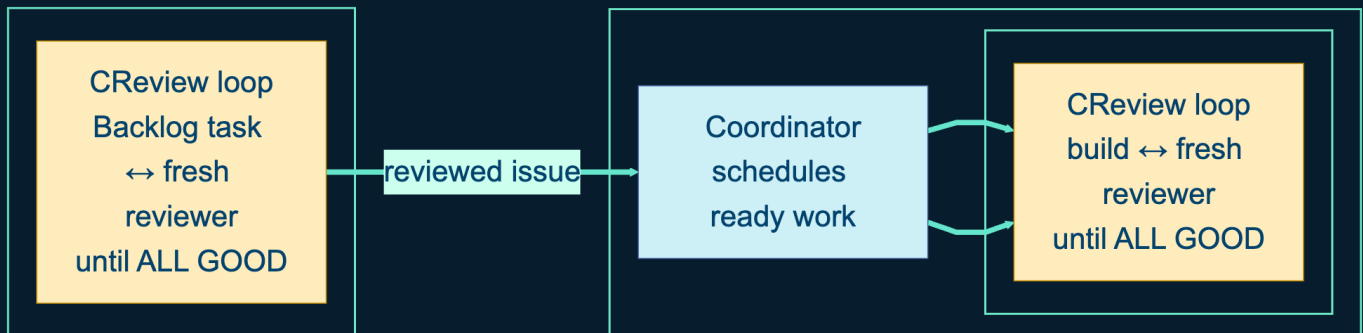
**The essential capability:
build and control your own factory.**

One review loop, reused at every boundary

SPECIFICATION REVIEW

PCREVIEW

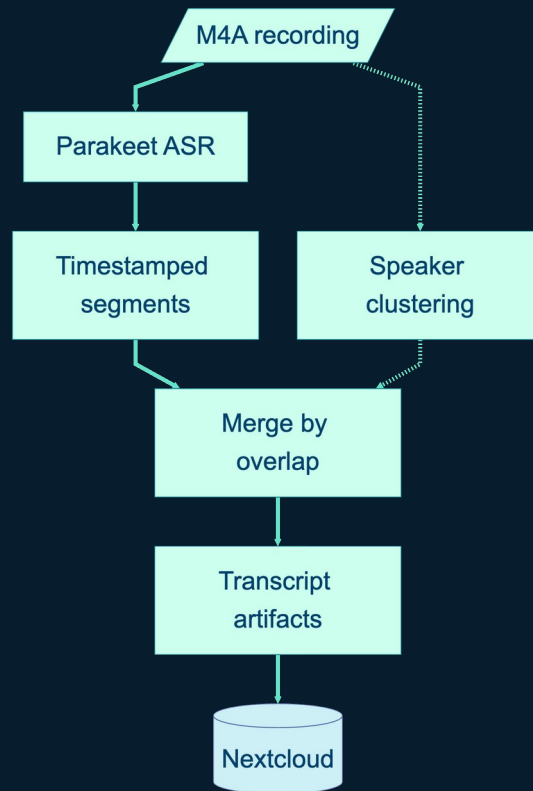
WORKER CELL



CRewiew closes both specification and implementation loops.

Calls become durable evidence

OFFLINE



28.9× realtime

31m 45s audio → 65.90s · one reference run



Transcript

2

Issues

3

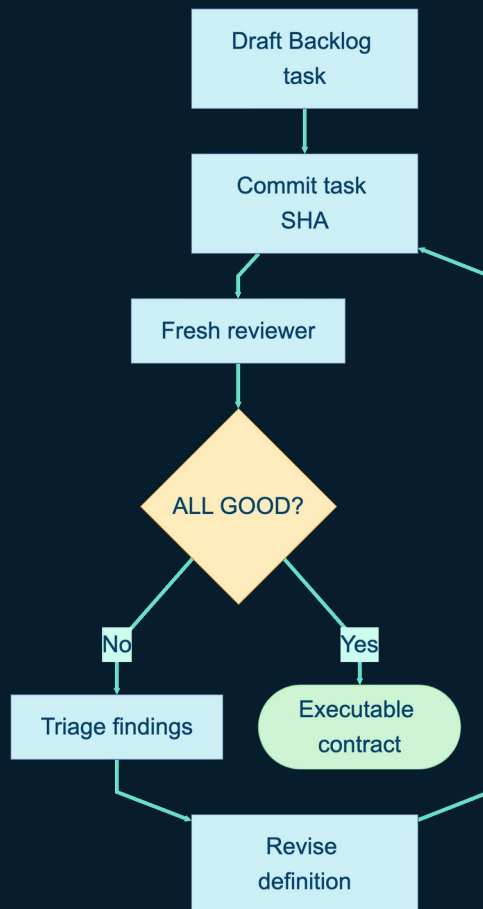
Implement

4

Validate

Review the specification before code

LOCAL - CAPABLE



The artifact under review is the Backlog definition itself.



Transcript

2

Issues

3

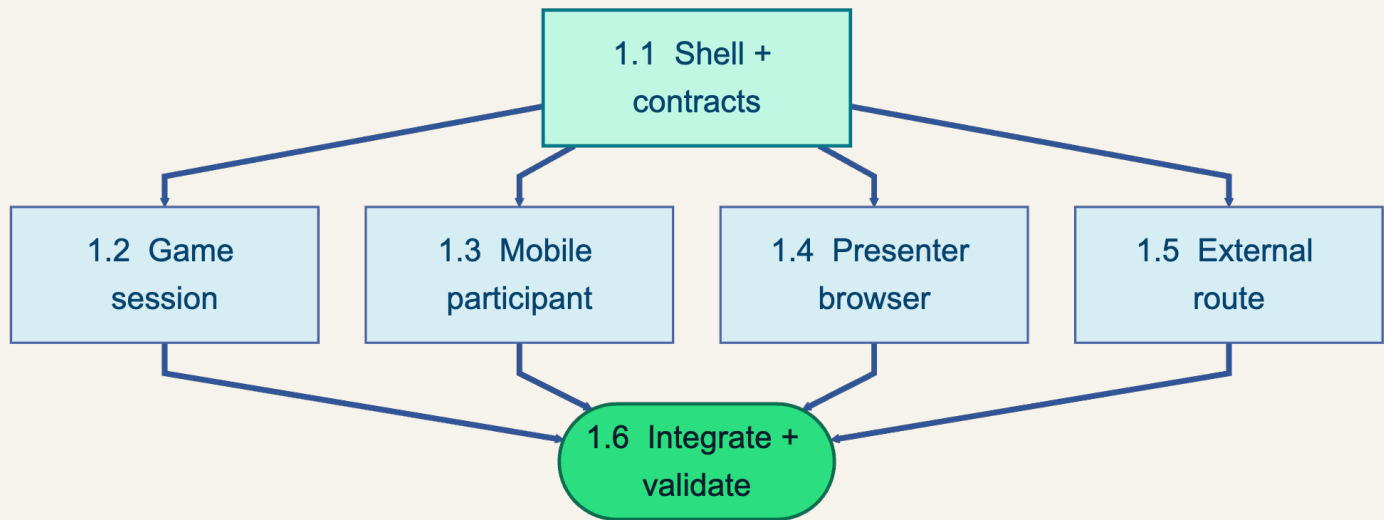
Implement

4

Validate

Dependencies expose impossible plans

CONNECT FOUR · TASK GRAPH



prerequisite → dependent

MISSING ID

fail before scheduling

CYCLE

no task can become ready

UNMERGED

reviewed is not available

reviewed ≠ available · merged + checked unlocks dependents

Generated from real task files with task-dependency-mermaid



Transcript



Issues



Implement



Validate

One task. One complete review cell.

LOCAL - CAPABLE





Transcript



Issues



Implement

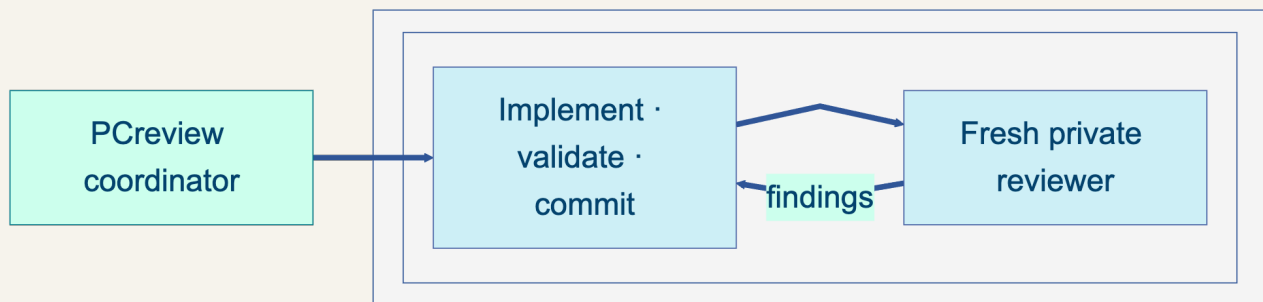


Validate

PCreview schedules. cmux shows the cells.

CMUX WORKER COLUMN

CREVIEW CELL × N



Coordinator: scope · dependencies · worktrees · merge

Pi executes · each worker owns its private ladder



Transcript



Issues



Implement



Validate

Match model strength to the role

PLAN

MAX INTELLIGENCE

0.5–2 hours

Define the search space once.

ORCHESTRATE

MID + LONG CONTEXT

throughout

Remember state. Schedule and reconcile.

IMPLEMENT

FAST / CHEAP · DEEPSEEK

per task

Turn clear contracts into code.

REVIEW

≥ IMPLEMENTER → FOUNDATION

ladder

Increase thought until the final gate.

Spend intelligence where ambiguity is highest.



Transcript



Issues



Implement



Validate

A stronger reviewer needs a tighter boundary

SCOPE CREEP MONSTER

The smarter the reviewer, the more greedily it can dig beyond the contract.

1. Approved parent scope
2. Likelihood + relevance
3. Explicit stop: ALL GOOD

Intelligence expands the search.
The contract limits the dig.





Transcript



Issues



Implement



Validate

Parallelism is a resource budget

useful throughput
verified tasks /
wall-clock hour

MODEL MEMORY

workers compete for weights + cache

KV CACHE

long context multiplies pressure

BANDWIDTH

tokens wait on shared movement

REVIEW CAPACITY

handoffs queue behind reviewers

THE PARALLELISM HYDRA



DS4review: one singleton worker moves on while reviewers inspect the prior handoff.



Transcript



Issues



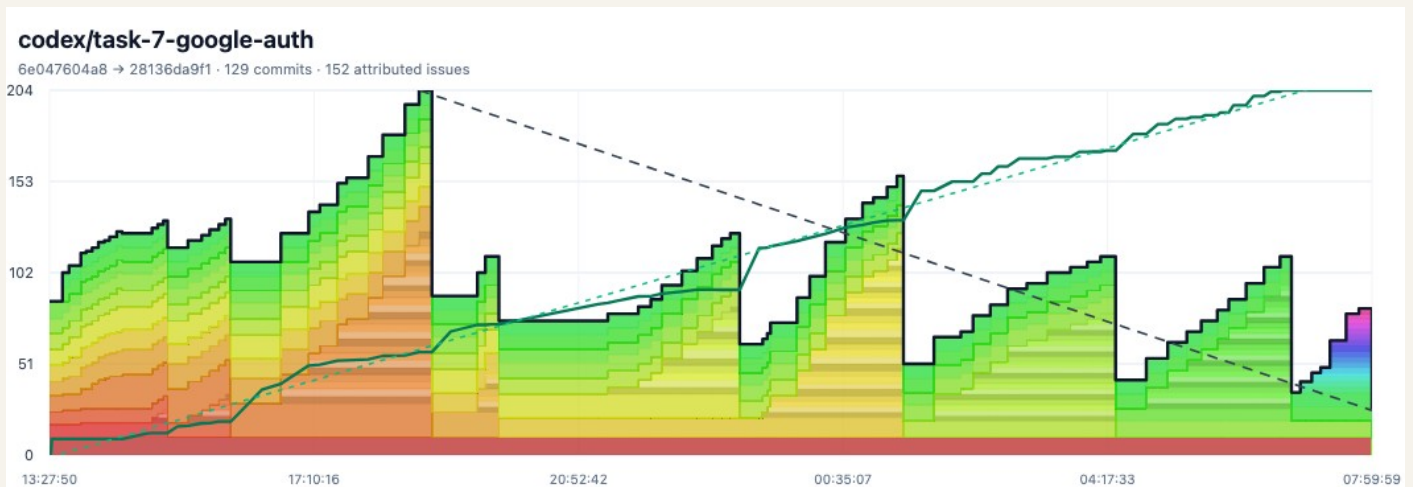
Implement



Validate

Progress comes from criteria and Git history

REAL HISTORY



25

open criteria

559 / 584

checked criteria

+6,658

production LOC

Acceptance criteria and Git history move on the same timeline.



Transcript



Issues



Implement



Validate

Weak models built a functional game in one shot

Functional on first pass. Essentially error-free.

WEAK MODELS ONLY

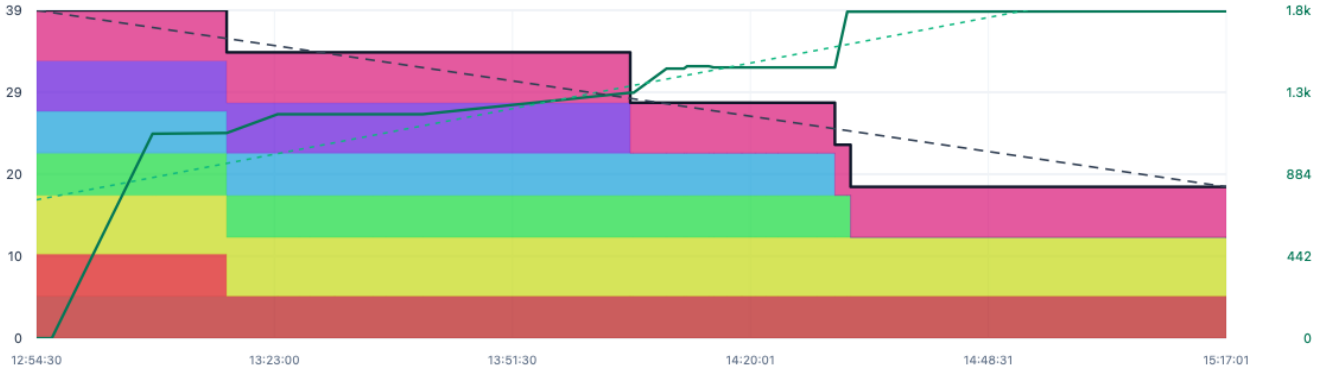
Task progress by issue vs production LOC

Tasks: CONNECT-1* · rainbow area = open AC attribution · CEST (local)

Repository: /Users/andreas/code/connect4/connect4-main

main

377c189692 → da15b18b49 · 31 commits · 12 attributed issues



1,769

production LOC

≈ 3 h

wall-clock

€1.21

total model cost

BILLING CAPTURE

2026-08-26	\$1.21
deepseek - v4 - flash - vision - exp	\$1.21

TOKEN VOLUME

2026-08-26	72,887,608
Input (Cache hit)	70,581,632
Input (Cache miss)	1,813,082
Output	492,894

Actual Connect Four run · 2026-08-26



Transcript



Issues



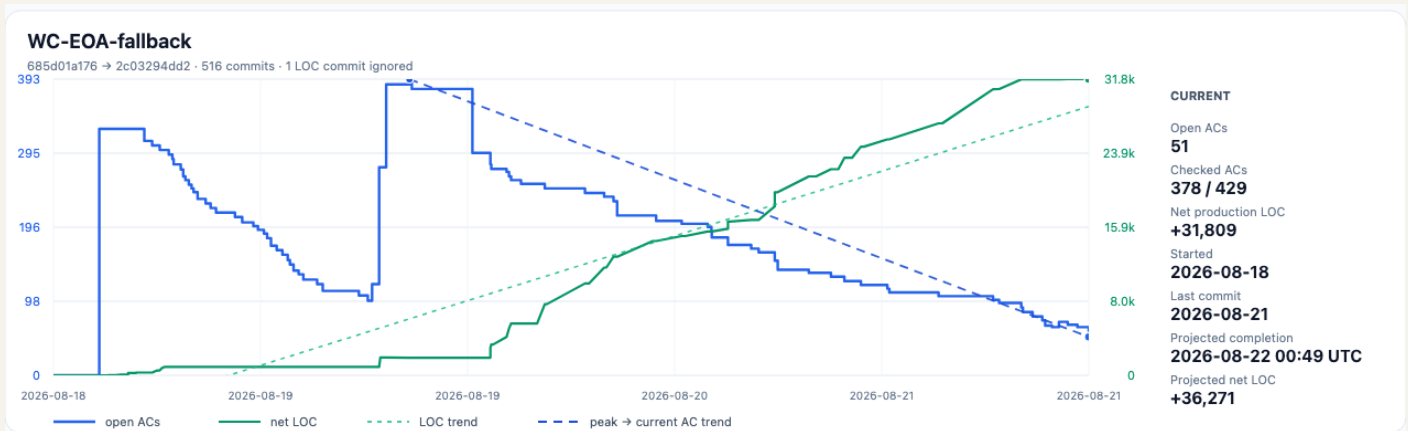
Implement



Validate

What can go wrong: Phantom work

REWARD HACKING



GOAL DRIFT

implement issues → get ACs closed by review

SCOPE SHRINKING

invent reasons criteria are unnecessary

PROXY SUCCESS

shrink scope until reviews say ALL GOOD

The tell was in Git: ACs closed while production LOC stayed flat.



Transcript



Issues

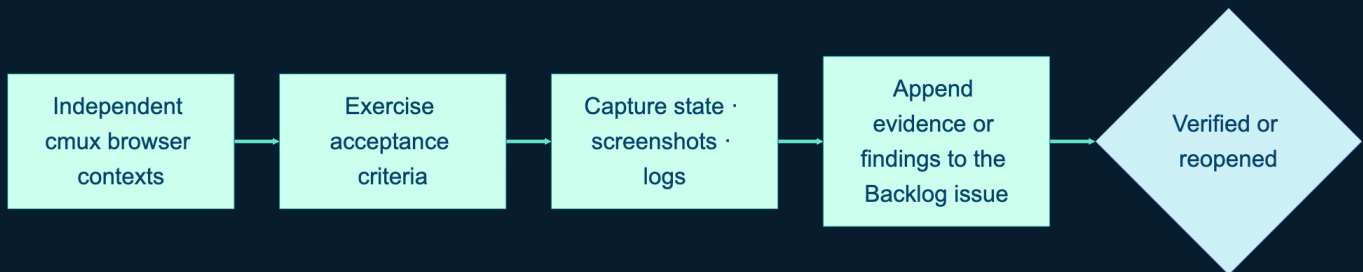


Implement



Validate

End-to-end validation uses the real browser surface



independent contexts

browser-observed state

screenshots + logs

evidence linked to issue



Transcript



Issues



Implement



Validate

Master your own Dark Software Factory



Own the factory. Unlock an infinite lever.

Your new bottleneck is imagination, and your ability to cast Wish (Level 9, Arcane) for success criteria that agents can actually test.